CCNA-12

Secure Campus Segmentation and Access Control

Zone-Based Access Policy, VLAN Architecture, and Multi-Area OSPFv2 Inter-Zone Routing for Fairmont Integrated Defense Solutions

Certification track: Cisco Certified Network Associate (CCNA 200-301) — Network Access, IP Connectivity, and Security Fundamentals, with CyberOps 200-201 security concepts as secondary alignment

Simulation platform: Cisco Packet Tracer 8.2 or later (no other software required)

Team size and effort: 4 students × 12 hours per week × 8 weeks = 384 total person-hours

PROJECT SCOPE DOCUMENT	
Project Name:	**Sponsor:**
Secure Campus Segmentation and Access Control Zone-Based Access Policy, VLAN Architecture, and Multi-Area OSPFv2 Inter-Zone Routing for Fairmont Integrated Defense Solutions	Fairmont Integrated Defense Solutions
Business Unit Project Leader:	**Project Manager:**
[Business Unit Project Leader]	Carl Edouard
Requested Start Date:	**Requested End Date:**
September 29, 2026	November 21, 2026
Objective:	
The goal of this project is to design, implement, and validate a zone-segmented campus network for Fairmont Integrated Defense Solutions (FIDS), in which inter-zone communication is governed by explicit access policy enforced at the routing layer, so that unauthorized lateral movement between network segments is architecturally prevented and demonstrably verifiable — every zone boundary proven to block what it should block and permit what it should permit, in both directions, with hit-counter evidence.	
Narrative:	
Fairmont Integrated Defense Solutions operates a single campus housing multiple departments with materially different data-sensitivity requirements. The existing flat network provides no enforced boundary between an engineering workstation handling proprietary schematics and a contractor terminal on the same physical infrastructure. Any device can attempt to reach any other, and the only barrier is policy documented in a spreadsheet rather than enforced by the network itself. For a defense-sector operator, a boundary that exists only on paper is not a boundary at all. This project replaces the flat architecture with a zone-segmented design in which each department occupies a dedicated VLAN and inter-zone communication is controlled by extended access lists applied at the routing layer. Five zones reflect real operational boundaries: Engineering, Corporate, Contractor/Guest, Server/DMZ, and a Management backbone. Engineering holds workstations processing sensitive project data and may be reached only from Corporate through an explicit permit policy. Contractor/Guest is internet-routed and categorically denied access to internal resources. The Server/DMZ hosts internal services reachable from Engineering and Corporate but not from Contractor. The Management zone forms the routing backbone and carries no end-user traffic.	

Multi-area OSPFv2 provides dynamic inter-zone routing within the management backbone, with each zone represented as its own area and the Management backbone as Area 0, so the design scales without manual route maintenance and the team practises the area-border behaviour that CCNA examines. Extended ACLs enforced at each zone boundary implement the access policy. Because Cisco IOS evaluates ACL entries in order, the team must reason explicitly about permit-and-deny sequencing — the single most common source of a silent policy gap, where connectivity appears to work but the wrong devices can communicate.

The intellectual core of the project is proving a negative. It is easy to show that a permitted flow works; the security value is in demonstrating that a denied flow is genuinely blocked, and in doing so for every zone relationship rather than a chosen few. The team therefore builds a complete directional policy matrix — every ordered pair of zones, permitted or denied, each with a stated justification — and verifies every cell of it, capturing ACL hit counters that increment on the deny as positive evidence that the control fired. A boundary that is asserted but never tested in the disallowed direction is treated as unproven.

The design is hardened and made resilient. Backbone OSPF adjacencies are authenticated so a rogue router cannot inject routes, user-facing interfaces are made passive so hello packets are not broadcast where no neighbour should exist, and device access is protected. The team then measures what happens under failure: when a zone link drops, OSPF must reconverge automatically and the policy must still hold afterward; when a boundary ACL is removed and reapplied, the policy must restore with no residual state leak. The deliverable is a segmented campus whose every boundary has been proven, in simulation, to enforce the documented policy — and whose behaviour under link failure and policy change has been measured rather than assumed.

Risks:

What are the potential challenges that this project may face?

ACL policy gap or misconfigured permit/deny order

Impact: Cisco IOS processes ACL entries sequentially; a permit placed above a deny allows traffic the design intends to block, and the error is silent — connectivity works, but the wrong devices can communicate. This is the single most dangerous failure mode of the project because it presents as success. ***Mitigation:*** Every ACL is reviewed line by line against the zone policy matrix before application. The complete directional verification in Appendix F requires the denied direction of every zone pair to be tested with hit-counter evidence, so a misordered rule is caught rather than assumed correct.

VLAN trunking misconfiguration isolating a zone entirely

Impact: A missing trunk VLAN or an incorrect native-VLAN assignment can silently drop all traffic for a zone without an obvious error, making the failure difficult to diagnose under time pressure. ***Mitigation:*** Trunk configuration and VLAN database consistency are verified at the Week 3 gate before any ACL work begins. The Week 3 gate requires demonstrated end-to-end connectivity within every zone before inter-zone policy is applied.

Addressing plan that cannot summarize at area borders

Impact: If subnets are allocated without regard to area boundaries, no contiguous summary is possible and the multi-area design loses one of its central benefits, forcing a re-address late in the project. ***Mitigation:*** The addressing plan is completed and reviewed against summary boundaries at the Week 2 design review, before any device is configured. The team must prove on paper that each zone's subnets fall inside a single summary prefix.

Silent policy gap from an incomplete verification set

Impact: Teams commonly verify the handful of flows they expect to matter and leave most zone relationships untested, so a permissive gap in an unexamined direction survives to the demonstration. ***Mitigation:*** The verification is exhaustive by construction: every ordered zone pair is a row in the directional matrix and every row must be evidenced. A relationship with no test is a defect, not an omission.

Uneven workload distribution

Impact: Security projects concentrate complexity in ACL design and zone-boundary configuration, and teams frequently allow one member to own the policy layer while others configure access switches. This fails the presentation requirement that any member can answer questions on any part of the build. ***Mitigation:*** Each

member owns one zone end-to-end — its VLAN configuration, ACL design, and acceptance testing. A mandatory Week 5 cross-training rotation requires each member to diagnose a fault deliberately introduced into another member's zone.

Loss of the working simulation file

Impact: A corrupted or overwritten .pkt file late in the project is unrecoverable and would end the project without a deliverable. ***Mitigation:*** Mandatory commit of the .pkt file and exported device configurations to the team repository at the end of every working session, confirmed in the weekly gate checklist.

Dependencies:

What are the tools, people, technical conditions or software/platforms that the project depends on?

- Cisco Packet Tracer 8.2 or later installed on all four team members' machines at an identical version
- Cisco Networking Academy access for CCNA 200-301 material covering VLANs, OSPFv2, and ACLs
- A shared version control repository (GitHub or equivalent) holding the .pkt file, exported device configurations, and documentation
- A diagramming tool for the topology deliverables — draw.io, Lucidchart, or Microsoft Visio
- A spreadsheet tool for the addressing plan, the zone policy matrix, and the acceptance test matrix
- A weekly team meeting of no less than two hours with all four members present
- Sponsor design review at the end of Week 2 to approve the zone architecture and addressing plan before configuration begins

Boundaries: (In Scope / Out of Scope)

What requirements/features are considered part of the scope to consider the project complete?

IN SCOPE

- A five-zone VLAN architecture: Engineering (Area 1), Corporate (Area 2), Contractor/Guest (Area 3), Server/DMZ (Area 4), and the Management backbone (Area 0)
- Multi-area OSPFv2 routing between zones within the management backbone, with explicit router IDs on every routing device
- A VLSM addressing plan engineered to support clean summarization at area borders
- A documented zone policy matrix defining, for every ordered zone pair, whether traffic is permitted or denied and the requirement that justifies it
- Extended ACLs applied at each zone boundary enforcing the documented policy matrix, with permit/deny ordering reasoned about explicitly
- Per-service granularity in the DMZ boundary rules — permitting specific services from permitted zones while denying the rest — rather than a single all-or-nothing permit
- Inter-VLAN routing at the distribution layer with 802.1Q trunking between access and distribution
- A lightweight internal service hosted in the Server/DMZ, reachable from permitted zones and blocked from denied zones
- Complete directional verification: every ordered zone pair tested, permitted flows succeeding and denied flows blocked, with ACL hit-counter evidence
- ACL logging on representative deny rules, demonstrating that a blocked attempt is observable
- Device access hardening, OSPF authentication on backbone adjacencies, and passive interfaces on user-facing segments
- A convergence and failure test matrix covering zone-link failures and ACL removal-and-reapplication, measured three times each
- Full device configuration export, logical and physical topology diagrams, and a design rationale document

What requirements are considered part of the next phase, the work of other teams, or not included in what is expected for this team to do?

OUT OF SCOPE

- BGP, EIGRP, IS-IS, or any routing protocol other than OSPFv2
- IPv6 addressing and OSPFv3, which belong to a separate team's project
- MPLS, SD-WAN, DMVPN, or any overlay or tunnelling technology
- IPsec encryption of inter-zone traffic, which belongs to a separate team's project
- Quality of Service classification, marking, or queuing
- Wireless infrastructure of any kind
- Firewall appliances, intrusion detection platforms, or security monitoring systems — enforcement in this project is by router ACL, not by a dedicated security appliance
- Physical hardware, live-endpoint validation, or any deployment outside the Cisco Packet Tracer simulation environment
- Centralised AAA, identity infrastructure, or role-based administrative access, which belong to a separate team's project
- Network management platforms or automated configuration tooling
- Ongoing operational support or maintenance after handover

Project Deliverables:

What are the key deliverables that are expected from the team to submit by the end of the semester?

D1. Cisco Packet Tracer simulation file (.pkt)
A single integrated file containing the complete five-zone campus topology, opening without error in Packet Tracer 8.2 or later, with every device named per the agreed convention and every configuration saved to startup-config.

D2. Zone policy matrix and addressing plan
A spreadsheet documenting every subnet, mask, VLAN assignment, and zone boundary, plus the complete directional policy matrix — every ordered zone pair marked permit or deny with a stated justification — and a written demonstration that the addressing plan summarizes cleanly at area borders.

D3. Network design rationale document
A written analysis justifying the zone architecture, the zone-boundary decisions, the router ID scheme, the ACL policy decisions, and the reasoning behind every permitted and denied inter-zone traffic flow.

D4. ACL policy validation evidence
Captured output demonstrating that each ACL permit and deny rule produces the expected result, with before-and-after hit counters showing policy enforcement in real time, and ACL logging output for representative deny rules.

D5. Complete directional verification matrix
Every ordered zone pair tested in the disallowed and allowed directions as applicable, each cell evidenced — the exhaustive proof that no unexamined relationship hides a permissive gap.

D6. Convergence and failure test matrix
A completed table covering zone-isolation failures, link failures, and ACL removal-and-reapplication scenarios, recording three timed measurements per scenario with timestamped evidence.

D7. Network topology diagrams
A physical diagram showing devices, interfaces, and cabling, and a logical diagram showing zones, VLANs, subnets, routing boundaries, and ACL enforcement points.

D8. Exported device configurations
Plain-text running configuration for every router and switch in the topology, committed to the repository and annotated to explain non-obvious configuration choices.

D9. Final report, presentation, and video demonstration
The consolidated project report, the Week 8 slide deck, and the recorded demonstration as specified in Appendix I.

Project Assumptions:

What standards/policies should be followed?

- IEEE 802.1Q governs VLAN trunking behaviour and is the reference for any disputed interpretation
- RFC 2328 (OSPF Version 2) governs routing protocol behaviour
- Cisco hierarchical network design principles — core, distribution, and access separation — inform the zone architecture
- The zone policy is default-deny between zones; every permitted flow is an explicit, justified entry in the policy matrix
- A consistent device naming convention is agreed in Week 1 and applied to every device, interface description, and diagram label
- All privileged access is password-protected, passwords are encrypted, and every configuration change is committed to version control in the same working session in which it is made

What are the things that the sponsor assumes the team should have or provide?

- A working simulation demonstrating every item in the in-scope list
- Evidence for every acceptance test, including tests that fail, with root-cause analysis
- Complete directional verification of the zone policy matrix, with the denied direction of every zone pair evidenced
- A video demonstration showing Packet Tracer CLI interaction and live permit/deny enforcement
- Attendance by all four members at the final presentation, each able to answer questions on any zone
- A repository commit history showing genuine contribution from all four members across the full eight weeks

What are the things that the team assumes the sponsor should provide?

- A design review at the end of Week 2 to approve the zone architecture and addressing plan before configuration begins
- The grading rubric in advance, as supplied in Appendix G
- Clarification of scope questions within a reasonable turnaround during the eight-week window
- Access to a faculty member or teaching assistant for technical escalation when the team is blocked for more than one working session

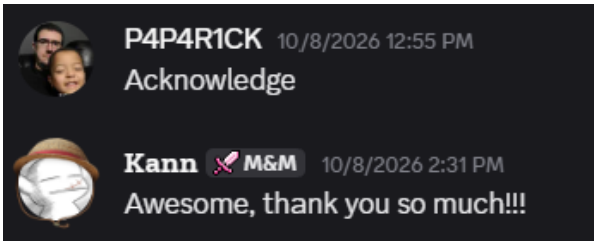
Estimated Time (and Budget):

The project will take 8 weeks to be completed, at 12 hours per week per student across a team of 4 students, for a total of 384 person-hours — matching the effort baseline of the approved capstone scope series. The Work Breakdown Structure in Appendix B allocates 48 person-hours per week and defines a completion gate for each week. Budget: $0. Cisco Packet Tracer 8.2 or later and Cisco Networking Academy course material are provided at no cost to enrolled students; no hardware purchase, cloud subscription, or commercial software licence is required to complete any deliverable in this scope.

Budget: $0. All tooling is free of charge — Cisco Packet Tracer 8.2 or later and Cisco Networking Academy course material are provided at no cost to enrolled students. No hardware purchase, cloud subscription, or commercial software licence is required to complete any deliverable in this scope.

Signatures

Sponsor Signature



Date: 10/13/2026

Project Manager Signature

Carl Edouard

Date: 10/13/2026

Appendix A — Cisco Certification and NetAcad Alignment

This project maps principally to the Network Access, IP Connectivity, and Security Fundamentals domains of CCNA 200-301, which together account for more than half of the exam. Because the design is fundamentally about enforcing a security boundary, the relevant CyberOps Associate 200-201 Security Concepts appear as secondary alignment — segmentation, least-privilege access policy, and defence-in-depth are demonstrated concretely rather than described. Every activity is tied to the NetAcad module that develops it.

Exam Domain	How This Project Covers It	NetAcad Reference
CCNA 2.1–2.2 — VLANs and interswitch connectivity	The five-zone VLAN architecture, 802.1Q trunking between access and distribution, and access-port assignment per zone	CCNA SRWE Module 3
CCNA 2.x — Inter-VLAN routing	Inter-VLAN routing at the distribution layer as the point where zone traffic is routed and where boundary ACLs are enforced	CCNA SRWE Module 4
CCNA 3.4 — Configure and verify OSPFv2	Multi-area OSPFv2 in the management backbone with explicit router IDs, area-border behaviour, and summarization at zone borders	CCNA ENSA Modules 1–2
CCNA 3.1 — Interpret the routing table	Reading and explaining route sources, administrative distance, and next-hop selection across zone boundaries	CCNA ENSA Module 1
CCNA 5.6 — Configure and verify access control lists	Extended ACL design and application at every zone boundary, per-service DMZ granularity, ACL logging, and permit/deny ordering verified by hit counters	CCNA ENSA Module 4
CCNA 1.6 — IPv4 addressing and subnetting	The VLSM plan engineered for zone segmentation and clean area summarization	CCNA ITN Modules 11–12
CCNA 5.x — Security fundamentals	Device access hardening, OSPF authentication on backbone adjacencies, and passive interfaces on user-facing segments	CCNA ENSA Module 3
CyberOps 1.x — Access control and least privilege (secondary)	The zone policy matrix as an access-control policy: default-deny between zones with explicit, justified permits, proven by testing the denied direction of every pair	CyberOps Associate Module 4
CyberOps 1.7 — Defence-in-depth (secondary)	Network segmentation as a defensive layer that architecturally prevents lateral movement, verified exhaustively across all zone relationships	CyberOps Associate Module 4

Appendix B — Eight-Week Work Breakdown Structure

Each week represents 48 person-hours (4 students × 12 hours). Hours shown are the planned team total for that week. The Gate column defines what must be demonstrably working before the team proceeds; a missed gate is the project manager's trigger to re-plan rather than to carry the deficit forward.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
1	Requirements, zone architecture, and design foundations	• Analyse the scope; produce a written requirements list and a sponsor question log • Study VLANs, OSPFv2, and ACLs through NetAcad SRWE Modules 3–4 and ENSA Modules 1–4 • Draft the physical topology: device placement, zone-to-area mapping, cabling plan • Agree the device naming and interface description convention • Create the repository and commit the initial topology skeleton	48	Physical topology and naming convention agreed by all four members; repository created with all members committing.
2	Addressing plan and zone policy matrix	• Build the complete VLSM addressing plan, engineered so each zone summarizes to a single prefix at its area border • Assign VLANs and map each zone to its OSPF area with written justification • Design the complete directional zone policy matrix — every ordered zone pair, permit or deny, with the requirement for each • Design the router ID scheme and the intended inter-zone routing paths before configuring anything • Present the addressing plan and policy matrix at the sponsor design review	48	SPONSOR DESIGN REVIEW. Addressing plan approved and the complete directional zone policy matrix signed off. No configuration begins until this gate passes.
3	Backbone and VLAN infrastructure	• Build the Management backbone and configure multi-area OSPFv2 with explicit router IDs • Bring up all VLANs and 802.1Q trunks; configure inter-VLAN routing at the distribution layer • Verify intra-zone connectivity in every zone and capture OSPF adjacency evidence • Confirm clean summarization at each area border against the Week 2 plan • Capture the pre-ACL baseline so policy enforcement can be shown as a change from it	48	All OSPFv2 adjacencies FULL; every zone reachable within itself; summarization visible at area borders. No ACLs applied yet.
4	Inter-zone routing and ACL enforcement	• Apply extended ACLs at every zone boundary per the policy matrix, reviewed line by line before application • Add per-service granularity to the DMZ boundary — permit specific services, deny the rest • Verify permitted inter-zone flows succeed and denied flows are blocked, in that order	48	Every permit and deny rule producing the expected result; ACL hit counters showing active enforcement.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
		• Capture before-and-after hit-counter evidence for each permit and deny rule • Confirm Contractor/Guest is categorically denied all internal resources		
5	Exhaustive policy verification and cross-training	• Execute the complete directional verification: every ordered zone pair, allowed and denied directions, each evidenced • Enable ACL logging on representative deny rules and confirm a blocked attempt is observable • Investigate and close any relationship that behaves differently from its policy-matrix entry • Record the full directional matrix with a hit-counter capture per cell • CROSS-TRAINING ROTATION: each member diagnoses a fault deliberately introduced into another member's zone	48	Every ordered zone pair in the directional matrix verified and evidenced; each member has diagnosed a fault outside their own zone.
6	Hardening and resilience	• Apply device access hardening and OSPF authentication on backbone adjacencies • Configure passive interfaces on all user-facing segments • Execute zone-link failure scenarios and measure OSPF convergence • Remove and reapply a boundary ACL; verify the policy restores with no residual state leak • Re-verify a sample of the directional matrix after reconvergence to confirm policy survives failure	48	All hardening applied; every failure scenario in the test matrix executed at least once; policy still enforced after each.
7	Full test execution and documentation	• Execute every acceptance test three times where timing applies • Record failover paths and convergence times with timestamped evidence • Export all device configurations and annotate non-obvious choices • Complete the logical and physical topology diagrams • Draft the final report and the design rationale document	48	Acceptance and convergence matrices complete with three measurements per timed scenario; all diagrams accurate and synchronized with the build.
8	Integration, rehearsal, and delivery	• Freeze the configuration; final regression run of the complete Appendix F test plan • Finalise the report including known limitations and root-cause analysis of any failed test • Record the video demonstration with all four members narrating • Build the slide deck and rehearse the presentation at least twice in full	48	FINAL DELIVERY. Full package submitted; all four members rehearsed and able to field questions on any part of the build.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
		• Mock question-and-answer session in which each member is questioned on a zone they did not own		

Appendix C — Team Roles and Hour Allocation

Four roles, 96 hours each. Roles describe accountability, not exclusive ownership: every student is expected to configure devices and to be able to explain any part of the build during the final presentation. The Project Manager role is a working role, not an administrative one — it carries a full technical workload in addition to coordination.

Role	Primary Accountability	Hrs
Project Manager and Backbone Owner — Carl Edouard	• Owns the Management zone and the multi-area OSPFv2 backbone configuration • Chairs the weekly integration meeting, owns the Week 2 design review, and maintains the Appendix B gate checklist • Owns the sponsor relationship and final authority on the addressing plan sign-off before configuration begins • Owns backbone OSPF authentication and the routing-domain hardening	96
Zone Policy and Addressing Lead — Jaden Torres	• Owns the VLSM addressing plan and the complete directional zone policy matrix as the single source of truth • Configures the Engineering zone end-to-end, including its VLAN, ACL, and acceptance testing • Designs and documents the extended-ACL policy for every zone boundary, reviewed line by line before application • Accountable for permit/deny ordering correctness and per-service DMZ granularity across all boundaries	96
Verification and Testing Lead — Michael Pietrewicz	• Owns the Corporate and Contractor/Guest zones end-to-end • Owns the complete directional verification matrix and the convergence and failure test matrix • Custodian of the Appendix F test log and the evidence archive • Accountable for ACL logging and for the exhaustiveness of the policy verification	96
Security, Documentation, and Diagram Lead — Devam Patel	• Owns the Server/DMZ zone, its hosted service, and all device access hardening • Produces and maintains both topology diagrams, keeping them synchronized with the build • Lead author of the final report and the design rationale document • Custodian of exported configurations in the repository	96

Collaboration and version control

- All four members commit to the shared repository; a commit history showing work from only one or two members is a deliverable failure regardless of whether the build works

- The .pkt file and exported configurations are committed at the end of every working session, so a corrupted file can be rolled back rather than rebuilt
- Because Packet Tracer files cannot be merged, only one member edits the master .pkt at a time; members build and test in their own copies and hand changes to the Project Manager for integration
- The addressing spreadsheet and the directional zone policy matrix are the single sources of truth; an address or policy rule changed on a device but not in the spreadsheet is treated as a defect
- Role assignments name Carl Edouard as Project Manager per the team's proposal; the remaining three zone-owner assignments are the team's to confirm and may be swapped without changing the scope
- The Week 5 cross-training rotation is mandatory — it is what makes the presentation requirement that any member can answer for any zone achievable

Appendix D — Topology Specification and Device Inventory

The topology models a single campus with five operational zones and a routed management backbone. Device models below are drawn from the standard Packet Tracer 8.2 palette; the team may substitute equivalents provided the required feature set is preserved and the substitution is documented. Quantities are minimums; the estate is sized so that a complete directional verification across all five zones is genuinely substantial rather than a handful of pings.

Qty	Packet Tracer Device	Role in the Topology
2	ISR 4331 Router	Management backbone routers running multi-area OSPFv2 (Area 0), forming the routed core between zones and carrying no end-user traffic
2	Catalyst 3560 Multilayer Switch	Distribution layer — inter-VLAN routing for all five zones and the enforcement point where extended boundary ACLs are applied; also the OSPF area-border between each zone area and Area 0
5	Catalyst 2960 Switch	Access-layer switches, one per zone (Engineering, Corporate, Contractor/Guest, Server/DMZ, Management), trunked to the distribution layer over 802.1Q
2	Server	Server/DMZ internal services reachable from Engineering and Corporate but denied to Contractor — consistent targets for the permit/deny acceptance tests, with per-service ACL granularity applied to them
10	PC	End devices distributed across the five zones — at least two per zone — providing the source and destination endpoints for the complete directional verification matrix

Design constraints the team must honour

- Each zone must occupy its own VLAN and its own OSPF area; the Management backbone is Area 0 and carries no end-user traffic
- Every routing device must have an explicitly configured router ID; inherited router IDs are a defect
- The zone policy must be default-deny between zones, with every permitted flow an explicit, justified entry in the policy matrix
- Every ACL must be reviewed line by line against the policy matrix before application, with permit/deny ordering reasoned about explicitly
- Every ordered zone pair must be verified: the permitted direction succeeds and the denied direction is attempted and blocked, with hit-counter evidence — an untested relationship is a defect
- Contractor/Guest must be categorically denied all internal resources and confirmed so from a host in that zone
- The DMZ boundary must apply per-service granularity — a single all-or-nothing permit to the DMZ does not satisfy the scope

- Every interface must carry a description identifying the far-end device and interface; no interface facing an end-user segment may send OSPF hello packets
- The addressing plan must summarize cleanly at every area border; the team must be able to show each zone as a single summary route

Appendix E — Addressing and Segmentation Plan

The address space is allocated so that each zone falls inside a contiguous prefix that summarizes cleanly at its area border. This spreadsheet is the single source of truth for the build; the team completes the per-VLAN host detail within each allocation and must preserve the summary boundaries, since a carelessly addressed zone cannot be summarized and the multi-area design loses its central benefit.

Zone	OSPF Area	VLAN	Allocation	Policy Posture
Engineering	Area 1	10	10.1.0.0/22	Sensitive. Reachable only from Corporate via explicit permit; denied from Contractor and, by default, all others.
Corporate	Area 2	20	10.2.0.0/22	May reach Engineering (permit) and Server/DMZ; source of the permitted inter-zone flows.
Contractor/Guest	Area 3	30	10.3.0.0/22	Internet-routed. Categorically denied all internal zones — the deny direction verified from a host here.
Server/DMZ	Area 4	40	10.4.0.0/22	Internal services reachable from Engineering and Corporate with per-service granularity; denied from Contractor. Hosts the acceptance-test targets.
Management backbone	Area 0	99	10.0.0.0/24	Routing backbone only; carries no end-user traffic. OSPF adjacencies authenticated.
Inter-zone / backbone links	Area 0	—	10.0.255.0/24	Point-to-point /30 or /31 links between backbone and distribution. Router IDs drawn from a systematic scheme.
Reserved for future zones	Areas 5+	—	10.5.0.0/16	Proves the design scales to further zones without re-addressing; one /22 per future zone.

Appendix F — Acceptance Criteria and Test Plan

This table is the definition of done. Each test must be executed and evidenced before the final presentation; the completed table, with evidence attached, is itself a graded deliverable. A test that cannot be made to pass must be documented as a known limitation with root-cause analysis — an honest, well-analysed failure scores higher than an undocumented one.

ID	Test Procedure	Expected Result	Evidence Required
T-01	Ping between all devices within each zone	100% success within every zone after ARP resolution; no packet loss	Screenshot per zone showing intra-zone reachability
T-02	Run 'show ip ospf neighbor' on every routing device	All expected adjacencies present and in FULL state; no unexpected neighbours	CLI capture from every routing device
T-03	Capture the routing table at a distribution switch and verify area summarization	Each zone appears as a summarized inter-area route as designed; routes trace to the intended next hops	'show ip route' capture with summary routes identified against the Week 2 design
T-04	Traceroute from Engineering to Corporate under normal conditions	The path matches the documented intended route from the Week 2 design	Traceroute output compared against the design document

ID	Test Procedure	Expected Result	Evidence Required
T-05	Attempt a connection from Corporate to Engineering (permitted)	Traffic permitted; the Engineering resource responds	CLI capture plus successful connection evidence, cross-referenced to the policy matrix
T-06	Attempt a connection from Corporate to a specific permitted service in Server/DMZ	The permitted service responds; a different, non-permitted service on the same host is denied	CLI capture showing the permitted service succeeding and the non-permitted service blocked — proof of per-service granularity
T-07	Attempt a connection from Contractor to Engineering (denied)	Traffic denied by the boundary ACL; the deny counter increments	CLI capture showing the ACL hit counter before and after the attempt
T-08	Attempt a connection from Contractor to Server/DMZ (denied)	Traffic denied by the boundary ACL	CLI capture showing the ACL hit counter incrementing on deny
T-09	Execute the complete directional verification matrix — every ordered zone pair	Every cell behaves exactly as its policy-matrix entry specifies; no relationship is left untested	The completed directional matrix with a hit-counter or connection capture per cell
T-10	Verify ACL logging on a representative deny rule	A blocked attempt produces an observable log entry identifying the denied flow	The log output captured alongside the attempt that produced it
T-11	Run 'show ip access-lists' on every ACL-bearing interface	All rules present in the correct order; hit counters reflect the test traffic	CLI capture per ACL-bearing device
T-12	Attempt to reach a restricted zone from Contractor after a plausible misconfiguration is deliberately introduced and then corrected	The gap is exposed while misconfigured and closed after correction — demonstrating the verification would catch a real error	Before-and-after captures across the introduced fault and its correction
T-13	Verify backbone OSPF adjacencies reject an unauthenticated neighbour	An adjacency without the authentication key is refused; no route injection is possible	CLI capture of the failed adjacency and the resulting log messages
T-14	Remove the ACL from one zone boundary, then reapply it, and verify the policy restores	Traffic is blocked again after reapplication with no persistent state leak	Before-and-after routing and ACL-table captures across the removal and reapplication
T-15	Shut down the primary link between two zones and measure convergence, three runs, then re-verify policy	OSPF reconverges and an alternate path installs automatically; the zone policy is still enforced after reconvergence	Timed ping output showing the loss window and recovery for each run, plus a post-reconvergence deny verification

Appendix G — Grading Rubric

Weights total 100%. The rubric is published with the scope so that teams can self-assess before submission.

Component	Wt	What full marks looks like
Topology build and completeness	15%	All five zones present and correctly interconnected; the .pkt opens without error; every device named per convention with full interface descriptions; configurations saved to startup-config.
OSPF and VLAN configuration correctness	20%	Multi-area OSPFv2 functioning in the backbone with all adjacencies FULL and explicit router IDs; VLAN trunking correct on all links; inter-VLAN routing working at the distribution layer; summarization demonstrated at area borders.

Component	Wt	What full marks looks like
ACL policy enforcement and exhaustive verification	30%	Every zone-boundary ACL producing correct permit and deny results with hit counters; permit/deny ordering correct; per-service DMZ granularity working; the complete directional matrix verified with no untested relationship and no policy gap.
Resilience and testing	15%	Convergence and failure matrix complete with three measurements per timed scenario; ACL removal-and-reapplication shows no residual state; policy re-verified and still enforced after reconvergence; OSPF authentication rejects an unauthenticated neighbour.
Design rationale and documentation	15%	Zone architecture and ACL policy justified on requirements; addressing plan supports segmentation and clean summarization; both diagrams accurate and synchronized with the final build; configurations exported and annotated.
Final presentation and video	5%	All four members present and able to answer questions on any zone; the live demonstration works; the video shows CLI interaction and live permit/deny enforcement.

Appendix H — Stretch Goals

The core scope in the table above is sized to occupy a competent four-student team for the full 384 hours. Stretch goals are strictly optional and are the mechanism by which a strong team distinguishes itself. No stretch goal may be attempted until every acceptance-criteria test in Appendix F passes — an incomplete core with an impressive stretch goal scores below a complete core alone.

- Add a physical proof-of-concept layer — a managed 802.1Q switch and a single-board-computer endpoint in the Contractor zone — that confirms the ACL policy blocks real traffic from a live device, aligning the physical addressing and VLANs with the simulation from the same authoritative plan (this was the team's original proposal and is preserved here as an optional extension for teams with access to hardware)
- Add a sixth zone for a partner organisation at a distinct trust level and design its inter-zone policy, analysing how the number of zone-pair rules grows as zones are added and what that means for policy maintainability
- Implement time-based ACLs so that Contractor access to a limited resource is permitted only during defined hours, and demonstrate the boundary opening and closing on schedule
- Introduce OSPF area-type optimization (stub or totally stubby) for a zone with no need for external routes, and prove the reduced routing table on that zone's router while policy enforcement is unaffected
- Build a written comparison of router-ACL enforcement against what a dedicated firewall would provide for the same five-zone policy, identifying which threats the ACL design cannot address and what visibility a firewall would add
- Produce a segmentation troubleshooting field guide mapping each failure symptom — zone unreachable, wrong device permitted, summarization not forming — to its diagnostic sequence and root cause

Appendix I — Final Presentation and Video Demonstration

Final presentation (Week 8)

A 10-minute presentation to the sponsor and faculty, followed by questions. All four students present. The panel will direct technical questions to individual students at random, so every member must be able to explain any part of the build — a team that can only answer through one spokesperson will be marked down.

- Business problem and design rationale — why this architecture, and what alternatives were rejected (3 min)
- Topology walkthrough against the live Packet Tracer file (5 min)
- Live demonstration of the acceptance criteria, including at least one deliberate failure/recovery scenario (7 min)
- Results, known limitations, and honest analysis of what did not work (3 min)
- Lessons learned and what the next phase would address (2 min)

Video demonstration

A recorded walkthrough of 8–12 minutes, submitted with the final package. The video must show live Packet Tracer interaction with visible CLI output — narrated slides alone are not acceptable. Every team member must narrate a section of the video.

- A walkthrough of the five-zone topology, identifying each zone, its VLAN, its OSPF area, and its boundary enforcement point
- Live 'show ip ospf neighbor' and a routing table showing summarized inter-area routes
- An ACL permit and deny demonstrated with hit counters incrementing in real time
- A walk through the complete directional matrix, showing that every zone relationship was verified in both directions
- A permitted service in the DMZ succeeding while a non-permitted service on the same host is denied — proof of per-service granularity

Final submission package

- The Packet Tracer .pkt file, opening without errors in version 8.2 or later
- Exported running configurations for every configured device, as plain text
- Network topology diagram, logical and physical views
- Completed Appendix F test plan with evidence attached
- Project report incorporating the design rationale, addressing plan, and known limitations
- Presentation slide deck
- Video demonstration file or link
- Version control repository link with a commit history showing contribution from all four students